電子學實驗 (一)
Integrators, Differentiators, and Their Frequency Responses
結報

B14901078 林孜謙、B14901153 沈威宇

1 Exp. 1 Integrator

I Circuit

Measure v_I and v_O waveforms of an op amp where input voltage v_I is connected to v_- through an input resistance $R = 1$ kΩ(current from v_I to v_-), v_+ is connected to ground, and v_O is fed back to v_- through a capacitance $C = 0.01$ μF (103) (current from v_- to v_O), under triangle, square, and sinusoidal v_I with amplitude 0.2.

II Theoretical results

$v_O(0^-)$ is the initial voltage across C (v_O relative to v_-).

Time domain:

$$v_+ = v_- = 0$$

$$i_I = \frac{v_I}{R} = i_C = -C\frac{\mathrm{d}v_O}{\mathrm{d}t}$$

$$v_O = v_O(0^-) - \frac{1}{RC}\int_0^t v_I(\tau)\,\mathrm{d}\tau$$

Laplace domain:

$$V_+ = V_- = 0$$

$$I_I = \frac{V_I}{R} = I_C = -CsV_O + Cv_O(0^-)$$

$$V_O = \frac{1}{s}v_O(0^-) - \frac{1}{RCs}V_I$$

Transfer function: Infinite DC gain. 90 degrees phase shift.

$$H(s) = \frac{V_o}{V_i} = -\frac{1}{RCs}$$

Unity-gain frequency

$$\frac{1}{2\pi RC} \approx 15.9 \text{ kHz}$$

Zero-state responses:

- Triangle wave input:

$$v_I(t) = 4A\left(\left|\frac{t}{T} - \left\lfloor\frac{t}{T} + \frac{1}{2}\right\rfloor\right| - \frac{1}{4}\right), \quad A, T > 0$$

$$v_O(t) = -\frac{AT}{RC}\left(2s\,|s| - s\right), \quad s = \frac{t}{T} - \left\lfloor\frac{t}{T} + \frac{1}{2}\right\rfloor$$

- Square wave input:

$$v_I(t) = 2A\left(\left\lfloor\frac{t}{T}\right\rfloor - \left\lfloor\frac{t}{T} + \frac{1}{2}\right\rfloor + \frac{1}{2}\right), \quad A, T > 0$$

  Triangle wave output with amplitude $\frac{AT}{4RC}$

$$v_O(t) = -\frac{AT}{RC}\left(\left|\frac{t}{T} - \left\lfloor\frac{t}{T} + \frac{1}{2}\right\rfloor\right| - \frac{1}{4}\right)$$

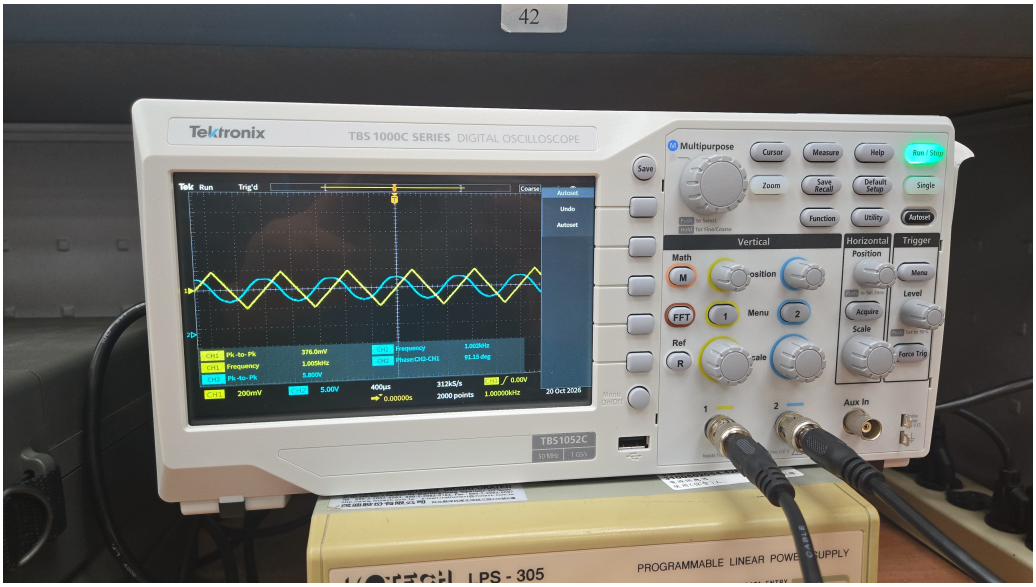
- Sinusoidal input:

$$v_I(t) = A\sin\left(\frac{t}{T}\right), \quad A, T > 0$$

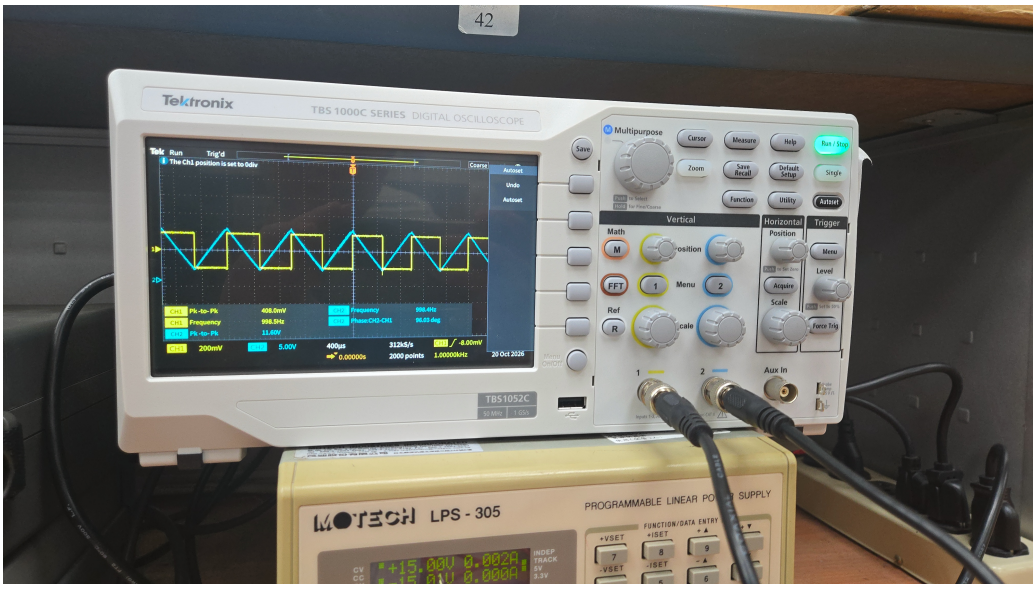
Sinusoidal output 90 degrees leading input with amplitude $\frac{AT}{RC}$.

$$v_O(t) = \frac{AT}{RC}\cos\left(\frac{t}{T}\right)$$

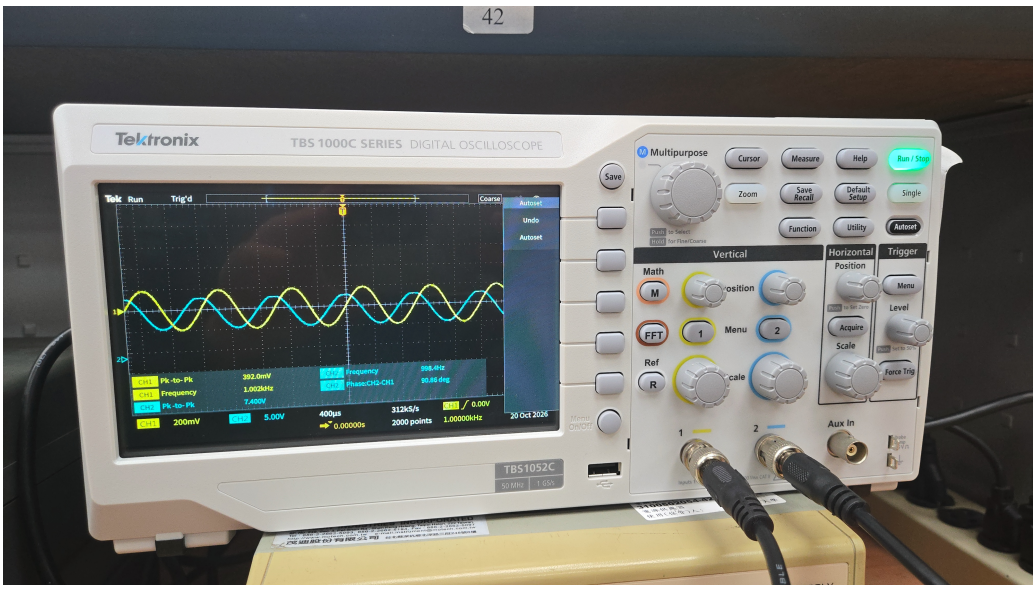
III 實驗結果與數據分析



Integrator triangle wave input



Integrator square wave input



Integrator sinusoidal wave input

All three agree with the theoretical results.

2 Exp. 2 Differentiator

I Circuit

Measure v_I and v_O waveforms of an op amp where input voltage v_I is connected to v_- through an input capacitor $C = 0.01$ µF (103) (current from v_I to v_-), v_+ is connected to ground, and v_O is fed back to v_- through a resistor $R = 1$ kΩ(current from v_- to v_O), under triangle, square, and sinusoidal v_I with amplitude 0.2.

II Theoretical results

$v_I(0^-)$ is the initial voltage across C (v_I relative to v_-).

Time domain:

$$i_I = C\frac{\mathrm{d}v_I}{\mathrm{d}t} = i_f = \frac{-v_O}{R}$$

$$v_O = -RC\frac{\mathrm{d}v_I}{\mathrm{d}t}$$

Laplace domain:

$$I_I = CsV_I - Cv_I(0^-) = I_f = \frac{-V_O}{R}$$

$$V_O = -RCsV_I + RCv_I(0^-)$$

Transfer function: Infinite gain at $\omega = \infty$. -90 degrees phase shift.

$$H(s) = \frac{V_o}{V_i} = -RCs$$

Unity-gain frequency

$$\frac{1}{2\pi RC} \approx 15.9 \text{ kHz}$$

Responses:

- Triangle wave input:

$$v_I(t) = 4A\left(\left|\frac{t}{T} - \left\lfloor\frac{t}{T} + \frac{1}{2}\right\rfloor\right| - \frac{1}{4}\right), \quad A, T > 0$$

  Square wave with amplitude $\frac{4ARC}{T}$

$$v_O(t) = -\frac{8ARC}{T}\left(\left\lfloor\frac{t}{T}\right\rfloor - \left\lfloor\frac{t}{T} + \frac{1}{2}\right\rfloor + \frac{1}{2}\right), \quad \frac{2t}{T} \notin \mathbb{Z}$$

- Square wave input:

$$v_I(t) = 2A\left(\left\lfloor\frac{t}{T}\right\rfloor - \left\lfloor\frac{t}{T} + \frac{1}{2}\right\rfloor + \frac{1}{2}\right), \quad A, T > 0$$

  Dirac delta sum output

$$v_O(t) = 2ARC\sum_{n\in\mathbb{Z}} \delta(t - nT) - \delta(t - \left(n + \frac{1}{2}\right)T)$$

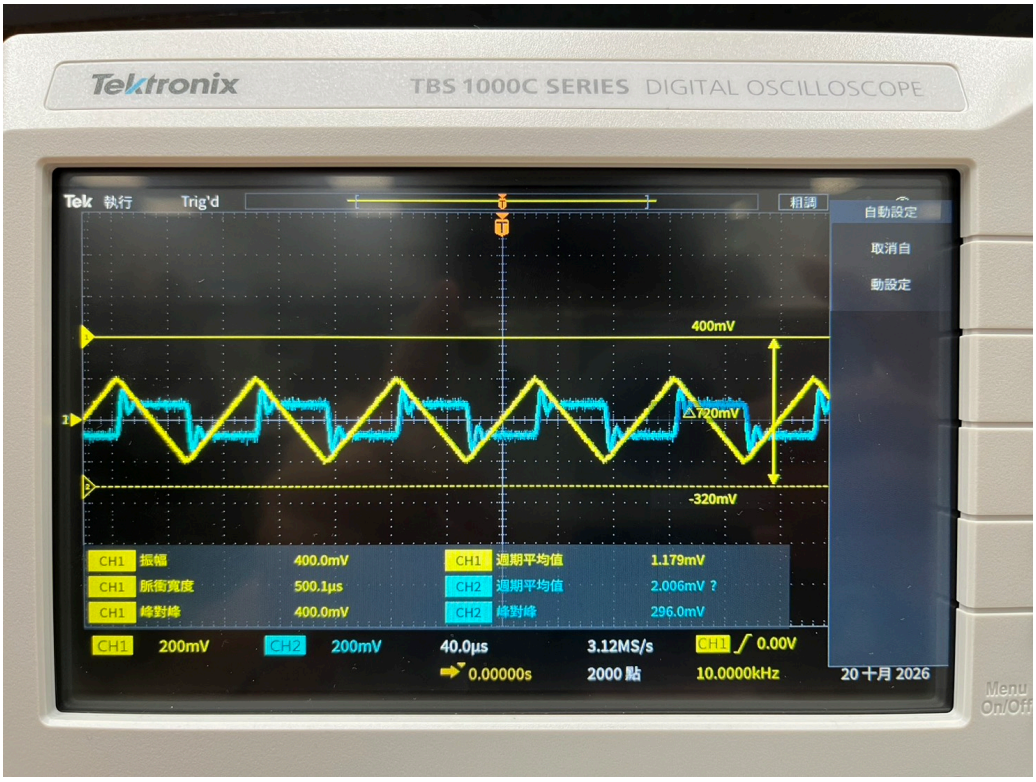
- Sinusoidal input:

$$v_I(t) = A\sin\left(\frac{t}{T}\right), \quad A, T > 0$$

  Sinusoidal output 90 degrees lagging input with amplitude $\frac{ARC}{T}$.

$$v_O(t) = -\frac{ARC}{T}\cos\left(\frac{t}{T}\right)$$

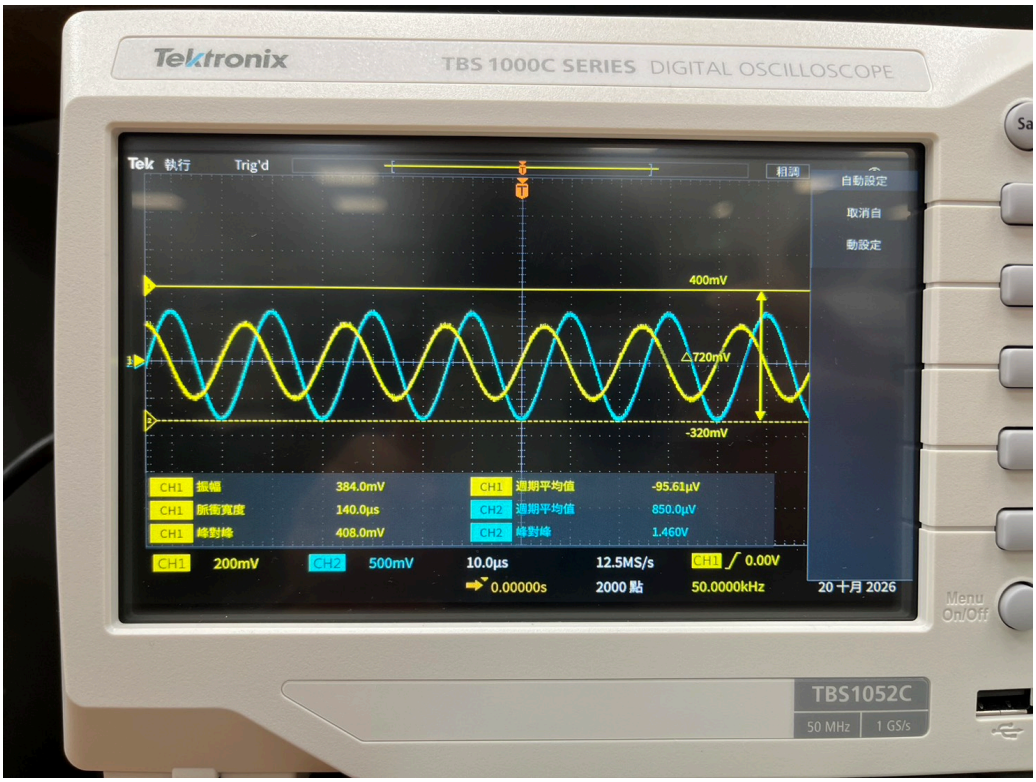
III　實驗結果與數據分析



Differentiator triangle wave input



Differentiator square wave input



Differentiator sinusoidal wave input

All three agree with the theoretical results. For the triangle wave input, the underdamping effect at each transition of square wave output is likely due to parasitic impedances.

3 Exp. 3 Frequency Responses (I) Integrator

I Circuit

Measure frequency response of the circuit in Exp. 1 Integrator under

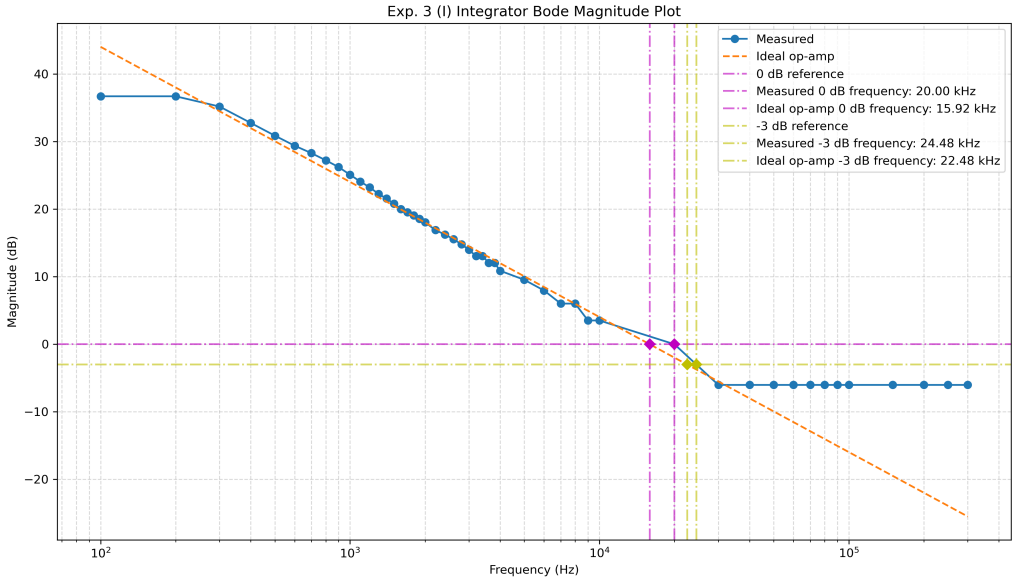
$$v_I = 0.2\sin(2\pi f t), \quad f = 100\text{ Hz} \sim 300\text{ kHz}$$

and sketch the Bode magnitude plot.

II　實驗結果與數據分析

frequency (Hz)	measured output voltage peak-to-peak (V)	measured gain (dB)	ideal output voltage peak-to-peak (V)	ideal gain (dB)	error w.r.t. ideal (%)
100	27.40	36.71	63.66	44.04	-56.96
200	27.40	36.71	31.83	38.02	-13.92
300	23.00	35.19	21.22	34.49	8.38
400	17.40	32.77	15.92	32.00	9.33
500	14.00	30.88	12.73	30.06	9.96
600	11.80	29.40	10.61	28.47	11.21
700	10.40	28.30	9.09	27.13	14.35
800	9.20	27.23	7.96	25.97	15.61
900	8.20	26.24	7.07	24.95	15.92
1000	7.20	25.11	6.37	24.04	13.10
1100	6.40	24.08	5.79	23.21	10.58
1200	5.80	23.23	5.31	22.45	9.33
1300	5.20	22.28	4.90	21.76	6.19
1400	4.80	21.58	4.55	21.11	5.56
1500	4.40	20.83	4.24	20.51	3.67
1600	4.00	20.00	3.98	19.95	0.53
1700	3.80	19.55	3.74	19.43	1.47
1800	3.60	19.08	3.54	18.93	1.79
1900	3.40	18.59	3.35	18.46	1.47
2000	3.20	18.06	3.18	18.02	0.53
2200	2.80	16.90	2.89	17.19	-3.24
2400	2.60	16.26	2.65	16.43	-1.98
2600	2.40	15.56	2.45	15.74	-1.98
2800	2.20	14.81	2.27	15.09	-3.24
3000	2.00	13.98	2.12	14.49	-5.75
3200	1.80	13.06	1.99	13.93	-9.52
3400	1.80	13.06	1.87	13.41	-3.87
3600	1.60	12.04	1.77	12.91	-9.52
3800	1.60	12.04	1.68	12.44	-4.50
4000	1.40	10.88	1.59	12.00	-12.04
5000	1.20	9.54	1.27	10.06	-5.75
6000	1.00	7.96	1.06	8.47	-5.75
7000	0.80	6.02	0.91	7.13	-12.04
8000	0.80	6.02	0.80	5.97	0.53
9000	0.60	3.52	0.71	4.95	-15.18
10000	0.60	3.52	0.64	4.04	-5.75
20000	0.40	0.00	0.32	-1.98	25.66
30000	0.20	-6.02	0.21	-5.51	-5.75
40000	0.20	-6.02	0.16	-8.00	25.66
50000	0.20	-6.02	0.13	-9.94	57.08
60000	0.20	-6.02	0.11	-11.53	88.50
70000	0.20	-6.02	0.09	-12.87	119.91
80000	0.20	-6.02	0.08	-14.03	151.33
90000	0.20	-6.02	0.07	-15.05	182.74
100000	0.20	-6.02	0.06	-15.96	214.16
150000	0.20	-6.02	0.04	-19.49	371.24
200000	0.20	-6.02	0.03	-21.98	528.32
250000	0.20	-6.02	0.03	-23.92	685.40
300000	0.20	-6.02	0.02	-25.51	842.48

III Bode plot



4 Exp. 3 Frequency Responses (I) Differentiator

I Circuit

Measure frequency response of the circuit in Exp. 2 Differentiator under

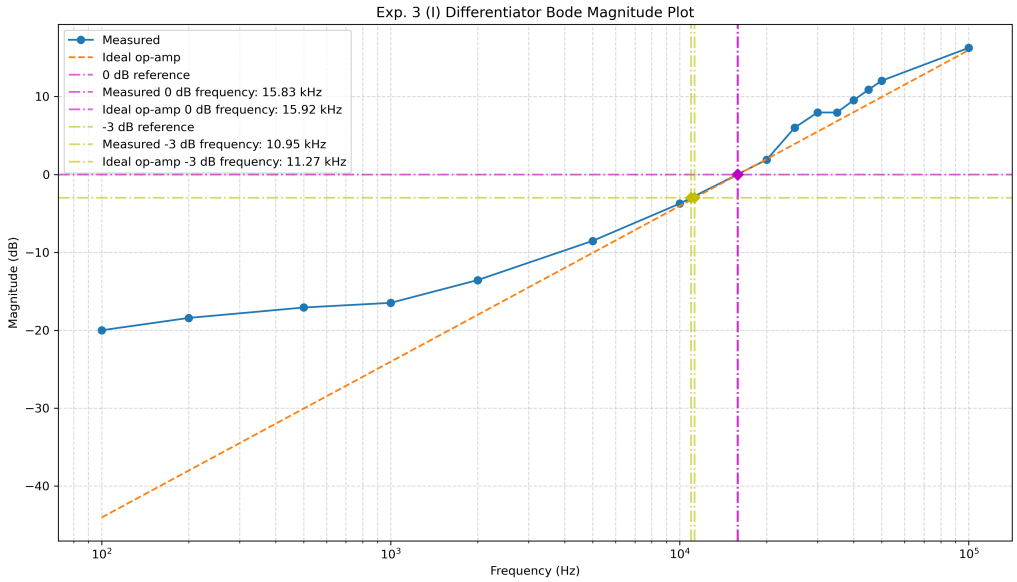
$$v_I = 0.2\sin(2\pi f t), \quad f = 100\text{ Hz} \sim 100\text{ kHz}$$

and sketch the Bode magnitude plot.

II 實驗結果與數據分析

frequency (Hz)	measured output voltage peak-to-peak (V)	measured gain (dB)	ideal output voltage peak-to-peak (V)	ideal gain (dB)	error w.r.t. ideal (%)
100	0.04	-20.00	0.00	-44.04	1491.55
200	0.05	-18.42	0.01	-38.02	854.93
500	0.06	-17.08	0.01	-30.06	345.63
1000	0.06	-16.48	0.03	-24.04	138.73
2000	0.08	-13.56	0.05	-18.02	67.11
5000	0.15	-8.52	0.13	-10.06	19.37
10000	0.26	-3.74	0.25	-4.04	3.45
20000	0.50	1.90	0.50	1.98	-0.93
25000	0.80	6.02	0.63	3.92	27.32
30000	1.00	7.96	0.75	5.51	32.63
35000	1.00	7.96	0.88	6.84	13.68
40000	1.20	9.54	1.01	8.00	19.37
45000	1.40	10.88	1.13	9.03	23.79
50000	1.60	12.04	1.26	9.94	27.32
100000	2.60	16.26	2.51	15.96	3.45

III Bode plot



5 Exp. 3 Frequency Responses (II) Integrator without feedback resistor

I Circuit

Measure frequency response of the circuit in Exp. 1 Integrator with the capacitor C replaced with a capacitor $C = 0.001$ μF (102) under

$$v_I = 0.2 \sin(2\pi f t), \quad f = 100 \text{ Hz} \sim 1 \text{ MHz}$$

and sketch the Bode magnitude plot.

II Theoretical results

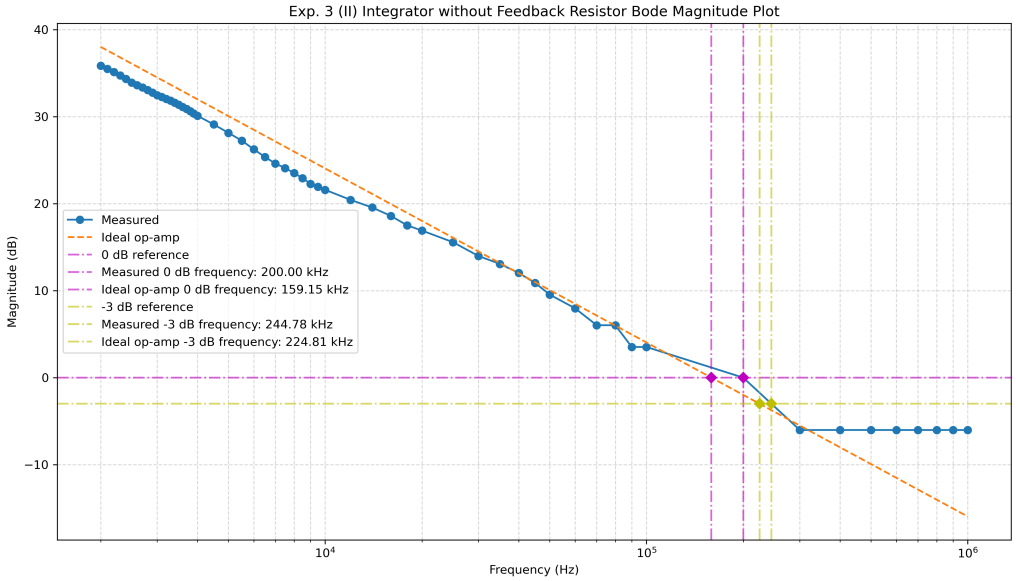
Unity-gain frequency

$$\frac{1}{2\pi RC} \approx 159 \text{ kHz}$$

III　實驗結果與數據分析

frequency (Hz)	measured output voltage peak-to-peak (V)	measured gain (dB)	ideal output voltage peak-to-peak (V)	ideal gain (dB)	error w.r.t. ideal (%)
2000	24.80	35.85	31.83	38.02	-22.09
2100	23.80	35.49	30.32	37.59	-21.49
2200	22.80	35.12	28.94	37.19	-21.21
2300	21.80	34.73	27.68	36.80	-21.24
2400	20.80	34.32	26.53	36.43	-21.59
2500	19.80	33.89	25.46	36.08	-22.25
2600	19.20	33.62	24.49	35.74	-21.59
2700	18.60	33.35	23.58	35.41	-21.11
2800	18.00	33.06	22.74	35.09	-20.83
2900	17.40	32.77	21.95	34.79	-20.74
3000	16.80	32.46	21.22	34.49	-20.83
3100	16.40	32.26	20.54	34.21	-20.14
3200	16.00	32.04	19.89	33.93	-19.58
3300	15.60	31.82	19.29	33.67	-19.14
3400	15.20	31.60	18.72	33.41	-18.82
3500	14.80	31.36	18.19	33.16	-18.63
3600	14.40	31.13	17.68	32.91	-18.57
3700	14.00	30.88	17.21	32.67	-18.63
3800	13.60	30.63	16.75	32.44	-18.82
3900	13.20	30.37	16.32	32.22	-19.14
4000	12.80	30.10	15.92	32.00	-19.58
4500	11.40	29.10	14.15	30.97	-19.42
5000	10.20	28.13	12.73	30.06	-19.89
5500	9.20	27.23	11.57	29.23	-20.52
6000	8.20	26.24	10.61	28.47	-22.72
6500	7.40	25.34	9.79	27.78	-24.44
7000	6.80	24.61	9.09	27.13	-25.23
7500	6.40	24.08	8.49	26.54	-24.60
8000	6.00	23.52	7.96	25.97	-24.60
8500	5.60	22.92	7.49	25.45	-25.23
9000	5.20	22.28	7.07	24.95	-26.49
9500	5.00	21.94	6.70	24.48	-25.39
10000	4.80	21.58	6.37	24.04	-24.60
12000	4.20	20.42	5.31	22.45	-20.83
14000	3.80	19.55	4.55	21.11	-16.43
16000	3.40	18.59	3.98	19.95	-14.55
18000	3.00	17.50	3.54	18.93	-15.18
20000	2.80	16.90	3.18	18.02	-12.04
25000	2.40	15.56	2.55	16.08	-5.75
30000	2.00	13.98	2.12	14.49	-5.75
35000	1.80	13.06	1.82	13.16	-1.04
40000	1.60	12.04	1.59	12.00	0.53
45000	1.40	10.88	1.41	10.97	-1.04
50000	1.20	9.54	1.27	10.06	-5.75
60000	1.00	7.96	1.06	8.47	-5.75
70000	0.80	6.02	0.91	7.13	-12.04
80000	0.80	6.02	0.80	5.97	0.53
90000	0.60	3.52	0.71	4.95	-15.18
100000	0.60	3.52	0.64	4.04	-5.75
200000	0.40	0.00	0.32	-1.98	25.66
300000	0.20	-6.02	0.21	-5.51	-5.75
400000	0.20	-6.02	0.16	-8.00	25.66
500000	0.20	-6.02	0.13	-9.94	57.08
600000	0.20	-6.02	0.11	-11.53	88.50
700000	0.20	-6.02	0.09	-12.87	119.91
800000	0.20	-6.02	0.08	-14.03	151.33
900000	0.20	-6.02	0.07	-15.05	182.74
1000000	0.20	-6.02	0.06	-15.96	214.16

IV Bode plot



6 Exp. 3 Frequency Responses (II) Integrator with feedback resistor

I Circuit

Measure frequency response of the circuit in Exp. 1 Integrator with the capacitor C replaced with a capacitor $C = 0.001$ μF (102) and a resistor R_F in parallel connection under

$$v_I = 0.2\sin(2\pi f t), \quad f = 100 \text{ Hz} \sim 1 \text{ MHz}$$

and sketch the Bode magnitude plot.

II Theoretical results

Laplace domain:

$$V_+ = V_- = 0$$

$$I_I = \frac{V_I}{R} = I_F = -\left(\frac{1}{R_F} + Cs\right)V_O + Cv_O(0^-) = -\frac{1 + R_F Cs}{R_F}V_O + Cv_O(0^-)$$

$$V_O = \frac{R_F C}{1 + R_F Cs}v_O(0^-) - \frac{\frac{R_F}{R}}{1 + R_F Cs}V_I$$

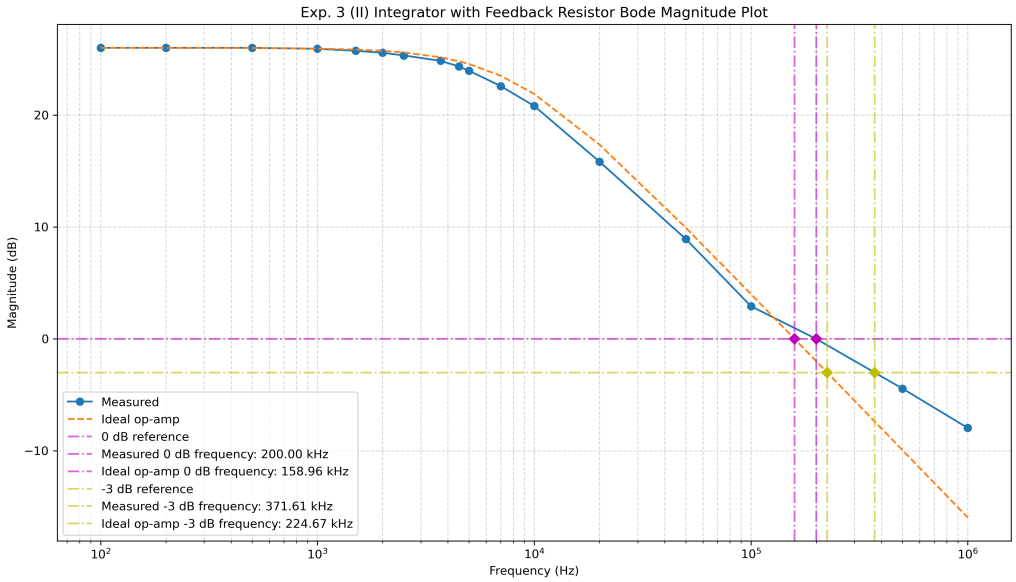
Transfer function: STC low-pass with cutoff frequency $\frac{1}{2\pi R_F C}$ and DC gain $\frac{R_F}{R}$.

$$H(s) = \frac{V_o}{V_i} = -\frac{\frac{R_F}{R}}{1 + R_F C s}$$

III 實驗結果與數據分析

frequency (Hz)	measured output voltage peak-to-peak (V)	measured gain (dB)	ideal output voltage peak-to-peak (V)	ideal gain (dB)	error w.r.t. ideal (%)
100	8.00	26.02	8.00	26.02	0.01
200	8.00	26.02	8.00	26.02	0.03
500	8.00	26.02	7.98	26.00	0.20
1000	7.92	25.93	7.94	25.95	-0.22
1500	7.76	25.76	7.86	25.87	-1.29
2000	7.60	25.58	7.76	25.75	-2.05
2500	7.40	25.34	7.63	25.61	-3.04
3700	7.00	24.86	7.25	25.17	-3.50
4500	6.60	24.35	6.96	24.82	-5.22
5000	6.32	23.97	6.77	24.58	-6.70
7000	5.40	22.61	6.01	23.53	-10.10
10000	4.40	20.83	4.98	21.91	-11.67
20000	2.48	15.85	2.96	17.38	-16.15
50000	1.12	8.94	1.26	9.95	-10.93
100000	0.56	2.92	0.63	4.01	-11.76
200000	0.40	0.00	0.32	-1.99	25.76
500000	0.24	-4.44	0.13	-9.94	88.52
1000000	0.16	-7.96	0.06	-15.96	151.34

IV Bode plot



7 誤差可能原因

The differences between the theoretical and experimental results may be attributed to the following factors:

- Measurement limitations: Oscilloscope has inadequate precision when the output amplitude becomes small.
- Non-ideal operational amplifier characteristics: The operational amplifier has finite open-loop gain, gain-bandwidth product, slew rate, etc.
- Output saturation: The ideal integrator has infinite DC gain. Small DC offsets and input bias may make low-frequency measurements deviate from the ideal op-amp model.
- Component tolerances: The actual resistance and capacitance values may differ from their nominal values, and the probes and contact points may have nonzero impedances.
- Parasitic effects: Parasitic impedances in the circuit and measurement equipment may affect the frequency response, especially at high frequencies.

8 實驗心得

I B14901078 林孜謙

PLACEHOLDER

II B14901153 沈威宇

Through this experiment, I gained a better understanding of integrators and differentiators, and realized the importance of avoiding infinite gain and accounting for practical limitations such as saturation and parasitic impedances in circuit design.

By measuring the gain at different frequencies, I observed deviations of the measured responses from the ideal predictions. For example, the integrator exhibited output saturation at low frequencies and reached the precision limit of the oscilloscope at high frequencies, while the differentiator showed oscillation near waveform transitions.

Plotting the measured and theoretical Bode plots and computing errors provided another useful perspective. It allowed me to compare their slopes, gain levels, and characteristic frequencies, while also enhancing my proficiency in using Python and Matplotlib.

Overall, this experiment showed me that designing a good circuit requires more than realizing the desired transfer function using ideal components. Dealing with practical

limitations is a more intriguing and difficult part that cannot be overemphasized.